

No.	Solution
1(a)	The Principle of moments states that, for a body in rotational equilibrium, the total clockwise moments about any point is equal to the total anti-clockwise moments about the same point.
1(b)(i)	The tension in the support at A, T_A is higher than the tension in the support at B, T_B. The total tension in the two support equal to the weight of the metal sign. Given that the ratio of the tension of the two supports is $\frac{3}{7}$, $\Rightarrow T_A = \frac{7}{10} \times 4.5 \times 9.81 = 30.9 \text{ N} \approx 31 \text{ N}$ $\Rightarrow T_B = 4.5 \times 9.81 - 30.9 = 13.2 \text{ N} \approx 13 \text{ N}$
1(b)(ii)	Let x be the perpendicular distance from point A to the weight of the metal sign. Taking moment about point A, Total clockwise moment = Total anticlockwise moment $4.5 \times 9.81 \times x = 13.2 \times 0.60$ $\Rightarrow x = 0.179 \text{ m}$ Hence horizontal distance $d = 0.179 + 0.20$ $= 0.379 \text{ m}$ $\approx 0.38 \text{ m}$
1(c)	The wind exerts a force on the metal sign to the right. To maintain equilibrium, there must be forces acting on the metal sign in opposite direction to the wind (to the left). These forces are the horizontal component of the forces at the hinge.
2(a)	Hooke's Law states that the extension (or compression) of a spring is directly